Answer **all** the questions in this section.

1 **(a)** Calcium carbide, CaC_2, is an ionic compound primarily used in industry to generate acetylene gas, C_2H_2, which has numerous applications in welding, cutting and chemical synthesis.

reaction 1 $CaC_2(s) + 2H_2O(l) \rightarrow Ca(OH)_2(s) + C_2H_2(g)$

(i) The carbon atoms in CaC_2 and C_2H_2 have sp hybridisation and hence they have similar bonding between the two carbon atoms.

Explain what is meant by sp hybridisation with reference to the carbon atoms in C_2H_2. **[2]**

(ii) Suggest how the electrons are arranged in the second shell of sp carbon atom so that the bonding in the two carbon atoms in C_2H_2 can occur. **[2]**

(iii) Explain, in terms of structure and bonding, why CaC_2 is a solid at room temperature and pressure while C_2H_2 is a gas. **[2]**

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(b) Calcium carbide reacts vigorously and explosively with water.

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Using the data in Table 1.1, calculate the standard enthalpy change of reaction for reaction 1.

Table 1.1

compound	$\Delta H_f^{\ominus}$ / kJ mol^{-1}
$CaC_2(s)$	– 59.0
$H_2O(l)$	–285.8
$Ca(OH)_2(s)$	– 985.2
$C_2H_2(g)$	+226.6

[2]

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(c) Caution is required when handling calcium carbide to minimise its exposure to water. Flammable C_2H_2 gas is produced upon reaction of calcium carbide with water and poses explosion risks.

(i) Define the term *standard enthalpy change of combustion, $\Delta H_c^{\ominus}$, of C_2H_2.* **[1]**

(ii) Use bond energy values from the *Data Booklet* to calculate the standard enthalpy change of combustion of C_2H_2. **[2]**

(iii) The actual standard enthalpy change of combustion of C_2H_2 is -1300 kJ mol^{-1}.

Other than the use of average bond energies, suggest **one other reason** for the difference between the actual value and the value calculated in **(c)(ii)**. **[1]**

(iv) Suggest an appropriate storage method for calcium carbide to minimise explosion risks. **[1]**

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[TURN OVER

(d) C_2H_2 is a weak Brønsted-Lowry acid and reacts with sodium amide, $NaNH_2$, to form an equilibrium mixture containing two acid-base pairs.

equilibrium 2 $\quad C_2H_2 + NaNH_2 \rightleftharpoons Na(C_2H) + NH_3 \quad K_c = 1 \times 10^{13}$

(i) With reference to equilibrium 2, explain the terms *Brønsted-Lowry acid* and *conjugate acid-base pair*. **[2]**

(ii) Identify another acid in the mixture in equilibrium 2 and suggest whether the acid is a stronger or weaker acid than C_2H_2. Explain your answer. **[2]**

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(e) C_2H_2 can undergo polymerisation in the presence of suitable catalysts to form polyacetylene, an organic polymer with alternating single and double bonds. It is used in organic semiconductors and its reaction with iodine produces highly electrical conducting material.

$$n\ HC{\equiv}CH \rightarrow -[CH{=}CH]_n-$$

$-[CH{=}CH]_n-$ is known as a repeating unit in polyacetylene.

(i) Two isomers can be obtained when C_2H_2 undergoes polymerisation.
Draw the structures of the two isomers, showing at least **two** repeating units, and state their isomeric relationship. **[2]**

Polyacetylene ($-[CH{=}CH]_n-$) becomes highly conductive when doped with iodine. The reaction removes an electron from a double bond, creating a radical cation called polaron.

$$-[CH{=}CH]_n- + \tfrac{3}{2}I_2 \rightarrow \underset{\text{polaron}}{-[CH{=}CH]_n^{\bullet +}} + I_3^-$$

(ii) State the role of iodine in the reaction. Explain your answer. **[2]**

(iii) Suggest why the doping process increases the electrical conductivity of polyacetylene. **[1]**

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